CHANG-YE TU

LANGUAGE

Mandarin Native

English TOEIC 945 (2019)

German Elementary

CODING

Python 15+ yrs

C/C++ 15+ yrs

Java 2 yrs

R 15+ yrs

MATLAB™ 15+ yrs

HTML/CSS/JS 15+ yrs

LaTeX 15+ yrs

CONTACT

chang-ye-tu.github.io

chang-ye-tu

RESEARCH

Sequential Decision Making under Uncertainty: Applications of

- Stochastic Optimal Control [9, 12, 5, 6, 7, 8]
- Theory of Optimal Stopping [1]
- Deep Learning and Reinforcement Learning [3]

Data-Driven Approaches to Sequential Decision Problems: [11]

Time-Inconsistent Control Theory and Its Applications: [10]

EDUCATION

PhD, Actuarial Science — Sep 2016 – Jul 2021

Department of Risk Management and Insurance
National Chengchi University, Taipei, Taiwan

Research topics:

- deep learning approach to option hedging [3]
- pricing and hedging problems of investment-linked insurance policies [4, 5]
- reinvestment risk of international bonds [1]
- optimal insurance regulatory scheme [12]

MSc, Applied Mechanics — Sep 1999 – Jun 2001

Institute of Applied Mechanics
National Taiwan University, Taipei, Taiwan

Research Topics:

- Stroh formalism in anisotropic elasticity
- numerical solution of boundary integral equation

BSc, Civil Engineering — Oct 1992 – Jun 1996

Department of Civil Engineering
National Taiwan University, Taipei, Taiwan

WORK EXPERIENCE

Project Assistant Professor — Feb 2022 – Current

Department of Mathematics
Soochow University, Taipei, Taiwan

Custom Software Developer — Mar 2005 – Sep 2016

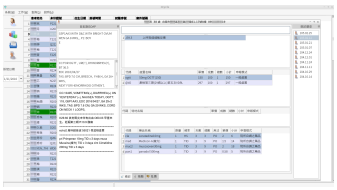
Independent

- Developed and maintained a clinical information system for Taiwan's National Health Insurance (NHI), successfully deployed across several clinics in Tainan; provided ongoing technical support and system optimization.
- Developed and maintained a custom ERP system for a manufacturing company using open-source technologies; implemented automated quality control through computer vision (OpenCV) and enhanced system security modules.

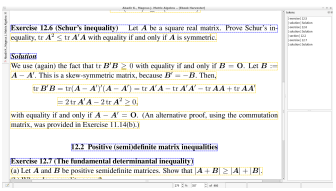
SOFTWARE

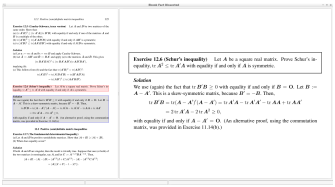
All software projects and papers with accompanying code are hosted on GitHub.

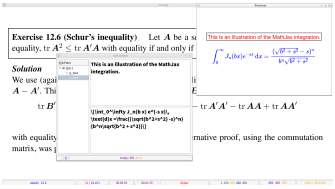
MYCIS: A clinical information system built exclusively for Taiwan's National Health Insurance (NHI)



Ānanda: A collection of adaptive computer assisted learning / memorizing subroutines.







Research Assistant Aug 2001 – Jul 2003

Department of Risk Management and Insurance
National Chengchi University, Taipei, Taiwan

- Conducted symbolic manipulation (application of Itô lemma) and numerical solution of nonlinear PDE by using Mathematica™ [8].
- Implemented Markov chain approximation method for numerically solving Hamilton-Jacobi-Bellman equations arising in asset allocation problem [6].

PUBLICATIONS

Journal Articles (peer reviewed)

[1] **C.-Y. Tu**† and S.-C. Chang, "Estimation of reinvestment risk of international bonds," *Review of Securities and Future Markets*, vol. 33, no. 4, pp. 77–102, 2021, (TSSCI).

[2] S.-C. Chang, Y.-K. Lee, W. Hsuan, and **C.-Y. Tu**, "Allocating overseas: Risk assessment of currency hedging in Taiwan life insurance industry," *Asia-Pacific Journal of Risk and Insurance*, vol. 14, no. 1, pp. 1–16, 2020, (EconLit).

[3] S.-C. Chang and **C.-Y. Tu**†, "Dynamic hedging of options by deep learning," *Insurance Monograph*, vol. 36, no. 4, pp. 1–20, 2020.

[4] **C.-Y. Tu**† and S.-C. Chang, "Numerical valuation of double barrier options: The finite element method approach," *Journal of Risk Management*, vol. 21, no. 1, pp. 5–21, 2019.

[5] S.-C. Chang and **C.-Y. Tu**†, "Optimal asset allocation under the liquidity constraint," *Journal of Risk Management*, vol. 20, no. 2, pp. 85–105, 2018.

[6] S.-C. Chang, **C.-Y. Tu**, and C.-H. Tsai, "Pension fund management using the Markov chain approximation," *Asia Pacific Management Review*, vol. 10, no. 4, pp. 259–266, 2005, (EconLit, TSSCI).

[7] S.-C. Chang, **C.-Y. Tu**, and Y.-S. Teng, "Speculating and hedging in optimal investment strategy for multi-period fund management," *Insurance Monograph*, vol. 19, no. 1, pp. 1–21, 2003.

[8] S.-C. Chang, C.-H. Tsai, C.-J. Tien, and **C.-Y. Tu**, "Dynamic funding and investment strategy for defined benefit pension schemes: Model incorporating asset-liability matching criterions," *Journal of Actuarial Practice*, vol. 10, pp. 131–155, 2002.

Preprints (submitted)

[9] W. Hsuan and **C.-Y. Tu**†, "On Merton's optimal consumption-investment problem: A Lie symmetry analysis approach."

Working Papers

[10] W. Hsuan and **C.-Y. Tu**†, "Insurance market equilibrium with catastrophic risk: A continuous-time mean-variance analysis."

[11] W. Hsuan and **C.-Y. Tu**†, "Quantifying reinvestment risk in international bonds: A random neural networks approach."

[12] **C.-Y. Tu**† and S.-C. Chang, "Optimal insurance solvency regulatory schemes under the early warning system."

AUTOBIOGRAPHY

I received my PhD in Actuarial Science from National Chengchi University. Previously I received my MSc in Applied Mechanics and BSc in Civil Engineering, both from National Taiwan University. Being initially trained in computational continuum mechanics and lately immersed in quantitative finance has equipped me with better understanding of the essence of interdisciplinary computing. Computing has been my greatest passion, and the aim of my research is to develop practical algorithmic solutions for real-world industrial problems.

The experience in scientific computing and quantitative finance coupled with decade long practices in computer-assisted learning/memorizing have shaped my thinking of effective learning and knowledge transfer. These experiences have built my confidence and an interest in teaching and I look forward to the opportunity to not only teach existing courses, but also work to develop new ones. I embrace every teaching opportunity, believe in keeping all my courses and tutoring sessions student-centered, and have worked enthusiastically and effectively with students at a variety of levels.